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A theoretical study for the performance evaluation of the two-dimensional carbon Kagome lattice as an anode material for lithium-ion batteries

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Securing high specific capacity in two-dimensional (2D) carbon allotrope materials in lithium-ion batteries (LIBs) is crucial for their use as well-balanced anode materials. Motivated by this, we utilize first-principles calculations to systematically investigate the potential of a carbon Kagome lattice (CKL) monolayer as an effective material for LIBs. A 2D CKL demonstrates excellent energetic, dynamic, and thermal stability. Lithium adsorption analysis demonstrates strong anchoring capability toward Li ions, characterized by a relatively high adsorption energy (−2.68 eV), arising from pronounced electron transfer (0.943 e) between Li ions and the CKL framework. Notably, the CKL presents a low Li diffusion barrier (0.18 eV), indicating fast ion transport kinetics that are highly desirable for high-rate battery operation. Furthermore, the maximum configuration of the CKL delivers a high theoretical capacity of 1115.7 mAh g^{−1}, simultaneously maintaining robust structural stability throughout Li-ion adsorption and desorption. Meanwhile, the calculated average open-circuit voltage is 1.3 V, which is comparable to that of TiO₂ (1.5–1.8 V) and h-AIC (1.38 V). Furthermore, the CKL still maintains good thermal stability even under the maximum Li storage capacity at 500 K. These merits demonstrate the CKL to be a promising and well-balanced carbon anode material and provide useful insights for the rational design of advanced anode materials for LIBs.

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1. Introduction

Rapid global transition toward renewable energy and the electrification of transportation have stimulated an unprecedented demand for high-performance energy storage systems. LIBs have greatly dominated this field due to their high energy density, long cycle life, and well-developed industrial infrastructure. Moreover, LIBs have been extensively employed in electric vehicles, portable electronic devices, and grid-scale energy storage systems.^{1–5} However, it has been demonstrated that Li-ion intercalation in conventional graphite anodes occurs predominantly at edge-plane sites, where the limited number of active regions results in sluggish diffusion kinetics and pronounced capacity fading, particularly under high-rate conditions. In addition, the theoretical specific capacity of graphite is limited to a relatively low value, and its interlayer

Li diffusion kinetics have approached the intrinsic limit, rendering it insufficient to meet the continuously increasing demands for higher energy density and fast-charging capability in applications.

To overcome the intrinsic energy density limitations of lithium-ion batteries, the development of advanced anode materials has become imperative. Two-dimensional anode materials have emerged as a promising strategy owing to their multiple stable adsorption sites,^{6–9} low diffusion barriers,^{10–12} and a large specific surface area.^{13–15} Since the successful synthesis of graphene,¹⁶ many researchers have considered it a potential material for LIBs. Although it exhibits remarkable electronic and mechanical properties, its extreme chemical inertness, resulting from the delocalized π -electron network, makes it unsuitable for use as an anode material in LIBs.¹⁷

Consequently, researchers have launched an intensive search for other 2D carbon allotropes, many of which have been theoretically predicted or experimentally synthesized.^{18–20} Furthermore, a recent study indicated that the introduction of non-hexagonal rings disrupts the delocalized π -electron system, leading to higher chemical activity and facilitating Li adsorption.²¹ For example, HOP-graphene (5–6–8 rings),²²

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β_{12} -borophene (3–6 rings),²³ χ -graphene (5–6–7 rings),²⁴ net- τ (4–5–6–8–10 rings),²⁵ ψ -graphene (5–6–7 rings),²⁶ penta-graphene (5 rings),²⁷ and Θ -graphene (5–6–8 rings)²⁸ exhibit high adsorption energy and low diffusion barriers, making them promising candidates for anode materials in LIBs. Nevertheless, to date, only a limited number of these two-dimensional carbon allotropes have been successfully synthesized in a controllable manner. Moreover, many 2D materials suffer from pronounced volume variation and unstable solid electrolyte interphase (SEI) formation during repeated charging and discharging processes, which severely hampers their cycling stability and practical applicability.⁷ Therefore, it is essential to explore novel two-dimensional carbon allotropes with improved stability and enhanced experimental accessibility.

In our study, a carbon Kagome lattice (CKL), which is based on the [110] surface of T-carbon,²⁹ consists of interconnected three-membered and twelve-membered carbon rings. Undoubtedly, such a non-hexagonal configuration breaks the perfect π -electron system inherent in graphene, which endows the CKL with higher chemical activity and renders it a highly promising candidate for lithium-ion battery anodes. The recent report on the properties of CKL has also attracted considerable attention.³⁰ Studies have found that CKL features a flat band near the Fermi level.³¹ Additionally, the periodic nanopores of CKL (porosity $\approx$ 97%, pore diameter $\approx$ 5.38 Å) are expected to provide efficient diffusion pathways for Li atoms.³¹ Importantly, the experimental realizability of CKL has already been demonstrated, as it can be fabricated *via* different synthesis strategies, such as picosecond-laser-induced transformation of carbon nanotubes in methanol and plasma-enhanced chemical vapor deposition on diamond substrates, thereby highlighting its practical viability.^{32,33} Overall, these insights into CKL frameworks justify and support a comprehensive evaluation of CKL as a feasible and efficient anode material for LIBs.

In this paper, first-principles density functional theory calculations are conducted to comprehensively evaluate the viability of the 2D CKL as an anode material for LIBs. We begin by establishing the CKL monolayer and assessing its energetic, dynamical, and thermal stability, along with its distinctive electronic structure characterized by a flat band near the Fermi level. Following this, Li adsorption is examined by gradually populating the energetically favorable sites, revealing robust binding strength and pronounced charge transfer. Building upon these insights, the electrochemical performance of CKL is further quantified, including its low Li diffusion barrier, suitable average open-circuit voltage, and exceptional Li storage capacity. Collectively, these results shed light on its potential application as an anode material in LIBs.

2. Computational methods

All first-principles calculations were carried out based on density functional theory (DFT) using the Vienna *ab initio* simulation package (VASP).³⁴ The crystal orbital Hamilton population (COHP) analysis was conducted using the LOBSTER

package.³⁵ The projector augmented-wave (PAW) method was employed to describe the electron-ion interactions. The exchange–correlation functional was treated within the generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) parameterization.³⁶ Considering the inherent underestimation of the band gaps using PBE, we used the Heyd–Scuseria–Ernzerhof (HSE06) hybrid functional for the recalculation of the band structure.³⁷ Besides, van der Waals (vdW) interactions were included using DFT-D3.³⁸ The phonon dispersion relations were obtained through the finite displacement method implemented in PHONOPY.³⁹ *Ab initio* molecular dynamics (AIMD) simulations were conducted in the NVT⁴⁰ ensemble using a $3 \times 3 \times 1$ supercell at 500 K and 800 K for 5 ps with a time step of 1 fs. To achieve accurate results while avoiding computational artifacts, a plane-wave basis set with a kinetic energy cutoff of 550 eV was used. A 20 Å vacuum layer was introduced along the z -axis to eliminate interactions between periodic images in the vertical direction. For structural optimization and density of states (DOS) calculations, Γ -centered k -point grids were employed, with grid sizes of $3 \times 3 \times 1$ and $10 \times 10 \times 1$ for different simulation setups. To explore Li-ion diffusion behavior, the climbing-image nudged elastic band (CI-NEB)⁴¹ method was adopted to determine the minimum energy paths and corresponding migration barriers between adjacent adsorption sites. Meanwhile, for the diffusion paths II and III, a single imaginary frequency was verified to confirm a true saddle point. The adsorption energy, formation energy, open-circuit voltage (OCV), and theoretical storage capacity were calculated from total-energy differences according to standard thermodynamic equations. Furthermore, the charge density difference (CDD) and Bader charge analyses⁴² were performed to investigate the charge redistribution and electron transfer mechanisms during Li adsorption.

3. Results and discussion

3.1. Crystal structure, stability, and electronic properties of CKL

The optimized crystal structure of the CKL monolayer is depicted in Fig. 1(a). The CKL structure, a planar sp^2 -hybridized 2D carbon allotrope cell, consists of six carbon atoms, with optimized lattice constants of $a = b = 5.19$ Å. Its framework features a mixed bonding configuration, where both C–C single bonds and C=C double bonds coexist. The structure primarily consists of interconnected three-membered and twelve-membered carbon rings. These structural features are in excellent agreement with previous studies.²⁹

To examine the dynamic stability of the material, we first calculated its phonon dispersion curves. Fig. 1(b) shows no significant imaginary modes throughout the Brillouin zone, except for negligible imaginary frequencies appearing near the Γ point, with amplitudes of about 0.86i THz. As noted by Wang *et al.*,⁴³ such small deviations are generally attributed to numerical artifacts in two-dimensional systems and do not affect the overall dynamic stability. Notably, the maximum

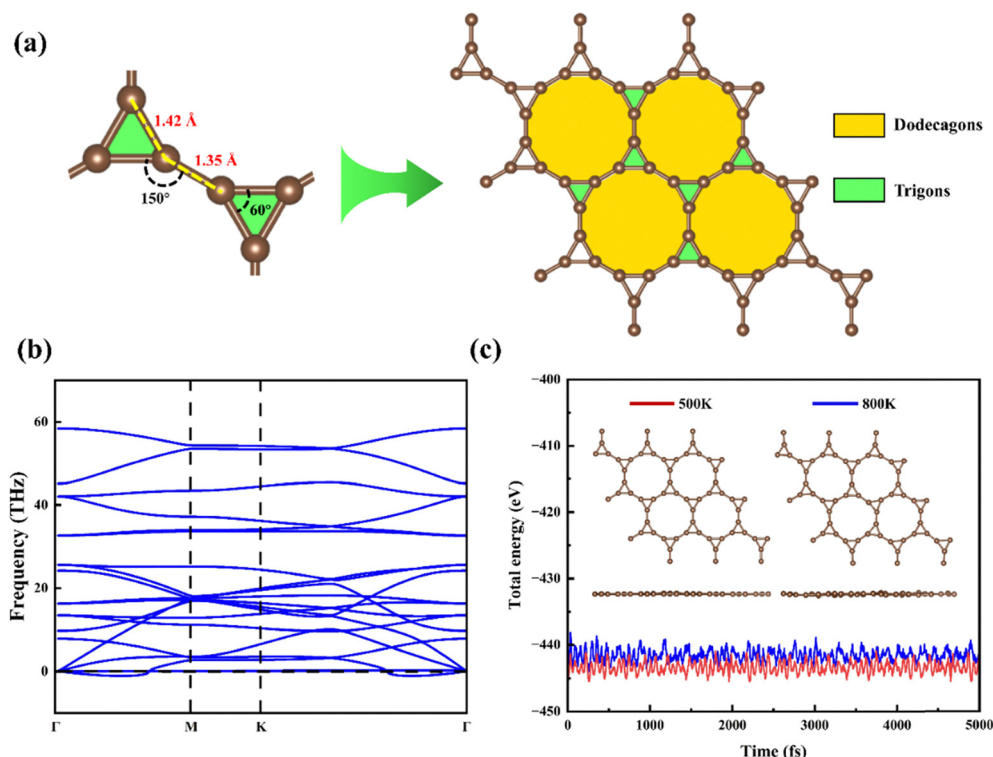


Fig. 1 (a) The unit cell and extended supercell of the CKL, emphasizing the periodic arrangement of dodecagons (yellow) and trigons (green). (b) Phonon spectrum along high-symmetry paths of the Brillouin zone. (c) Evolution of the total energy and snapshot of CKL during an AIMD simulation at 500 K and 800 K.

phonon frequency reaches 58.5 THz, which is higher than those of Me-graphene (47.4 THz)⁴⁴ and graphene (49.5 THz),⁴⁵ indicating the robustness and stability of the monolayer CKL. Furthermore, AIMD for 5 ps was also carried out at 500 K and 800 K to evaluate the thermal stability. As can be seen from Fig. 1(c), the structure reaches apparent thermal equilibrium within the first 2 ps, and then the structural framework maintains its structural integrity with minimal potential energy fluctuations during the remaining simulation time. Moreover, to further examine the stability of CKL over a longer total timesteps, an extended AIMD simulation with a total duration of 12 ps at 500 K was also conducted (Fig. S1), supporting the above analysis of 5 ps and demonstrating the robust thermal stability of the CKL.

The electronic band structure within the PBE functional and DOS of pristine CKL are shown in Fig. 2(a) and (b). The multiple bands crossing the Fermi level indicate that CKL exhibits metallic behavior. However, given that the PBE functional underestimates bandgaps, we recalculate the band structures using the HSE06 functional. Apparently, some systematic shifts of bands can be observed, and bands are also seen to cross the Fermi level, revealing that CKL is metallic. Analysis of the DOS reveals that the prominent peak near the Fermi level primarily originates from the C-pz orbitals. Furthermore, the electron localization function (ELF) was employed to explore the bonding characteristics in more detail. As displayed in Fig. 2(c), high ELF values are observed along the in-plane C–C bonds,

confirming the existence of robust covalent bonds. When the slice is moved approximately 1 Å above the atomic plane, the ELF values approach 0.5, indicating that a delocalized π -electron cloud is formed by the pz orbitals of carbon atoms, which contributes to the metallic nature of CKL. These research findings indicate the excellent electrical conductivity of 2D carbon materials. The unique structure, good stability, and inherent metallic properties of the CKL monolayer suggest its potential as an anode material for LIBs.

3.2. Adsorption of a single Li atom on the CKL

The stable adsorption of Li atoms on the CKL monolayer surface constitutes a fundamental requirement for its feasibility as an anode material in lithium-ion batteries. The interaction between Li and the CKL monolayer was therefore systematically investigated by considering five symmetrically equivalent sites, including two hollow sites (H1 and H2), two bridge sites (B1 and B2), and one top site (T1), as illustrated in Fig. 3(a). After full structural relaxation, only three stable adsorption configurations (H1, H2, and B1) were retained. Notably, Li atoms initially placed at the B2 or T1 sites spontaneously migrated to the nearby H2 site, indicating that these positions correspond to unstable adsorption states. This relaxation behavior clearly demonstrates that hollow sites are energetically preferred for Li adsorption on the CKL monolayer. Consequently, all subsequent analyses focus exclusively on hollow-site configurations. The adsorption energy (E_{ads}) of a

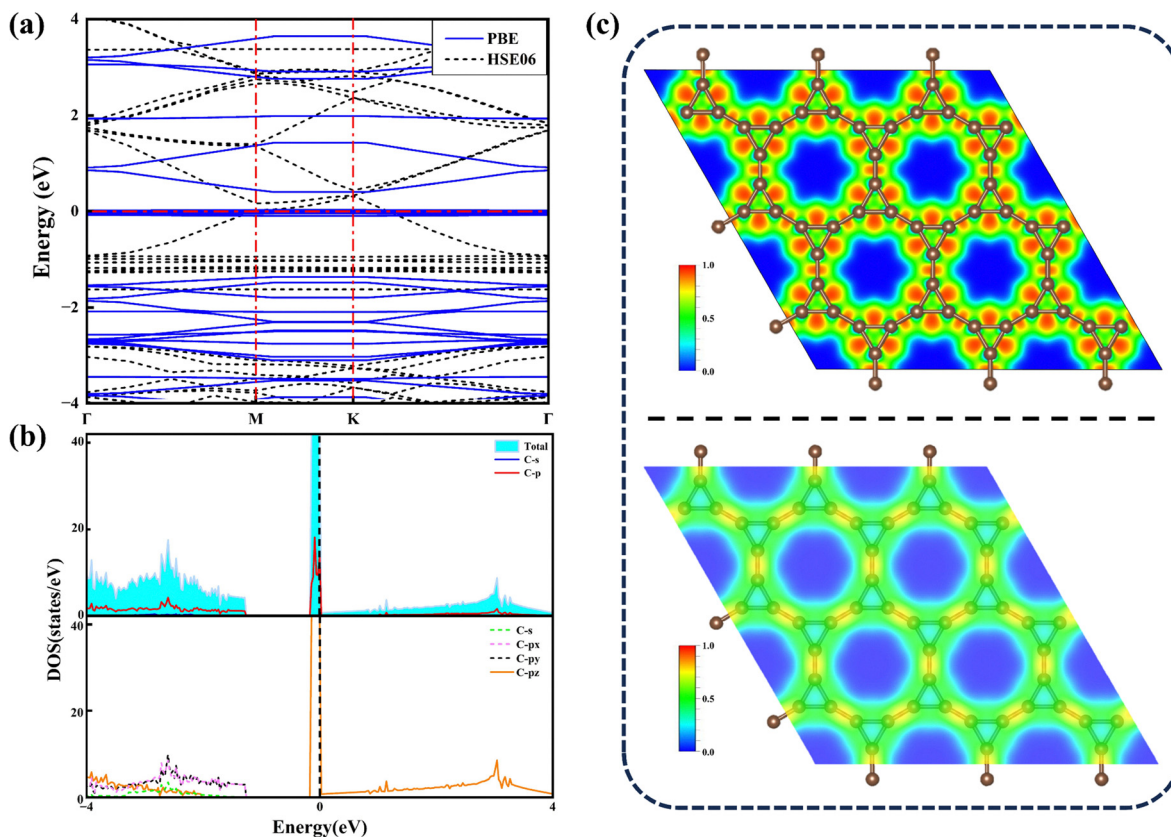


Fig. 2 (a) The band structure of CKL given by PBE and HSE06 functionals. (b) Orbit-projected density of states of CKL. The Fermi level is aligned to be 0 eV. (c) The electron localization function maps of CKL with slices crossing the structure plane and 1 Å above the structure plane.

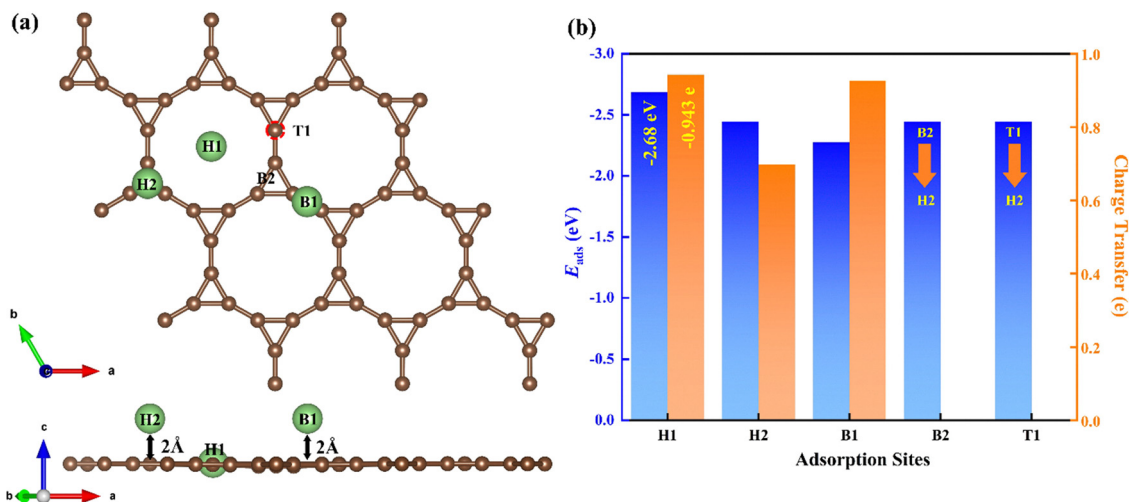


Fig. 3 (a) All the non-equivalent adsorption sites in CKL (H1, H2, B1, B2, and T1). (b) Adsorption energy and corresponding charge transfer of Li at different sites.

single Li atom at different sites on the CKL monolayer is determined using

$$E_{\text{ads}} = E_{\text{Li+CKL}} - E_{\text{CKL}} - E_{\text{Li}} \quad (1)$$

where $E_{\text{Li+CKL}}$ and E_{CKL} represent the total energies of the original CKL and after the adsorption of the Li atom,

respectively, and E_{Li} is the energy of a single Li atom. According to the definition of adsorption energy, negative adsorption energies imply that Li adsorption on the CKL monolayer is energetically favorable. Moreover, increasingly negative values reflect stronger interfacial interactions, corresponding to enhanced stability of the adsorbed system. As shown by the

computational results in Fig. 3(b), the most favorable adsorption site is identified as H1, where the Li atom is adsorbed parallel to the CKL plane with the lowest adsorption energy of -2.68 eV. The H2 and B1 sites exhibit slightly higher adsorption energies of -2.45 eV and -2.27 eV, with corresponding adsorption heights of 2.00 Å and 1.94 Å, respectively. The H1 site is located at the center of the 12-membered ring, where the relatively large internal angles generate a spacious cavity that effectively accommodates the Li atom, thereby ensuring enhanced adsorption stability.⁴⁶ The adsorption energies at all considered sites exceed the cohesive energy of bulk Li (-1.63 eV),⁴⁷ indicating that subsequent Li atoms energetically favour surface adsorption on CKL rather than aggregating into Li clusters.⁴⁸ Such a uniform Li distribution is expected to effectively suppress dendrite formation during lithiation, which is highly desirable for the practical application of high-performance LIBs.

To clarify the underlying mechanism of Li adsorption on CKL, we investigated the electronic properties of the system. As depicted in Fig. 4(a), the CKL inherently exhibits metallic behavior, and upon adsorption of a single Li atom, the density of states near the Fermi level increases. This enhancement is attributed to the electron donation from Li to the CKL substrate, which induces electrostatic interactions across the material, further boosting its conductivity. Upon further inspection of the PDOS for individual atoms, it is clear that the increase in states at the Fermi level originates primarily from the C-pz orbitals of the CKL, whereas the states of the metal Li show a loss, indicating that the metal atom donates electrons to the CKL substrate. In the energy range of 1.6 and 6.2 eV, the Li orbital exhibits insignificant fluctuation peaks, indicating weak orbital hybridization.

The bonding characteristics of Li–C interactions were elucidated by COHP and integrated COHP (ICOHP) analyses.

As shown in Fig. S2, weak bonding contributions are observed below the Fermi level, while pronounced antibonding features appear near and above it. Furthermore, the calculated ICOHP (-0.114) indicates negligible orbital interaction and a weak noncovalent interaction.

To further probe the charge transfer mechanism between Li and CKL, the charge density difference was calculated (see Fig. 4(b)), using the following expression:

$$\Delta\rho = \rho_{\text{Li+CKL}} - \rho_{\text{CKL}} - \rho_{\text{Li}} \quad (2)$$

where $\rho_{\text{Li+CKL}}$, ρ_{CKL} , and ρ_{Li} represent the charge densities of the Li-adsorbed system, the pristine surface, and an isolated Li atom, respectively. The CDD plots indicate that the Li atom donates $0.943e$ to the CKL surface. This strong ionic interaction between Li and the CKL surface contributes to the stability of the Li adsorbed structure and facilitates efficient electron transport.

3.3. Diffusion of a single Li atom on a CKL monolayer

Lithium-ion mobility on the electrode surface is crucial for the rate performance of LIBs. Therefore, we calculated the diffusion energy barrier for a single Li atom on monolayer CKL, considering three representative diffusion pathways: path I (H1–H1), path II (H1–H2), and path III (H2–H2). The corresponding diffusion barriers for each path are presented in Fig. 5. Our results indicate that the diffusion barriers are 0.30 eV for path I, 0.28 eV for path II, and 0.18 eV for path III. For path I, the Li atom was initially expected to migrate directly between two equivalent H1 sites by crossing the bridge region. However, during structural relaxation, the Li atom deviated from the designed trajectory and spontaneously passed through the neighboring H2 site before reaching the next H1 position. This change indicates that the H2 site acts as an intermediate

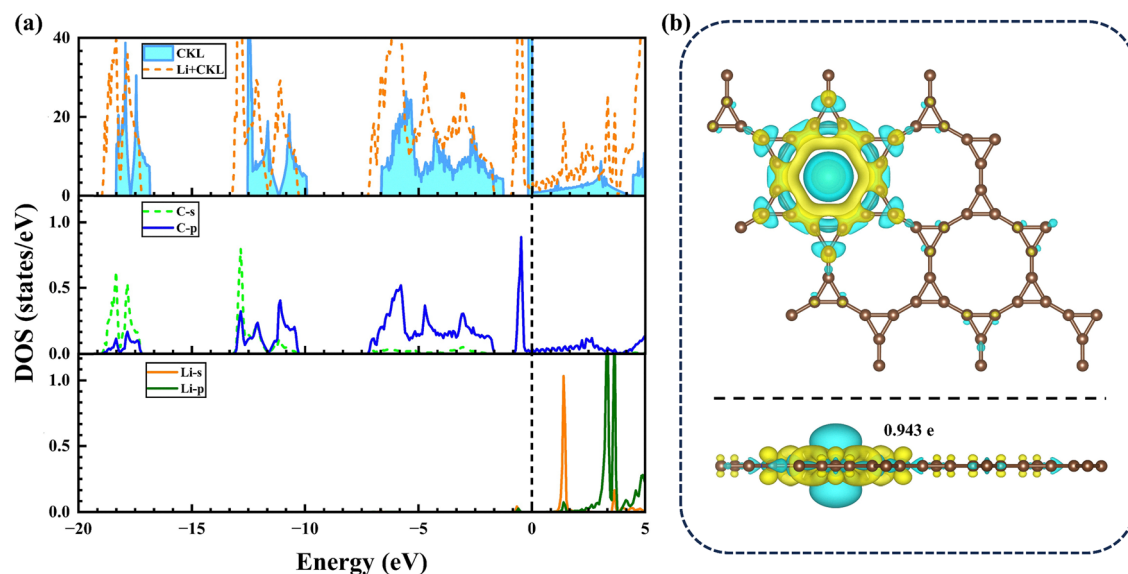


Fig. 4 (a) TDOS of pristine CKL and Li-adsorbed CKL, orbital-projected density of states for C atoms in CKL upon Li adsorption and orbital-projected density of states for the Li atom adsorbed on the CKL. (b) Top and side views of CDD for the Li atom adsorbed on the H1 site, the yellow and cyan regions correspond to charge accumulation and depletion, respectively, with an iso-value of $0.0012 \text{ e } \text{\AA}^{-3}$.

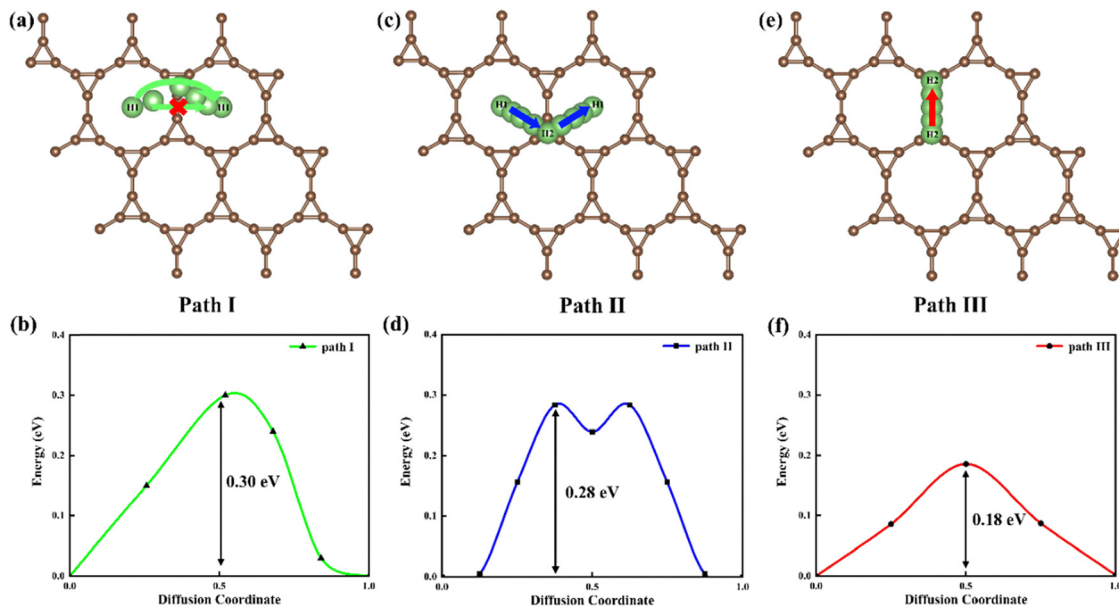


Fig. 5 The possible Li atom diffusion pathway and diffusion barrier on CKL along (a) and (b) path I, (c) and (d) path II, (e) and (f) path III.

metastable point along the diffusion route, forming a more energetically favorable transition channel. By contrast, the lower diffusion barrier along path II suggests that Li atoms preferentially diffuse along this pathway, ensuring a fast ion transport capability. The obtained diffusion barriers are comparable to or even much lower than those reported for other 2D carbon allotropes such as net- τ (0.30–0.36 eV),⁴⁹ M-graphene (0.34–0.37 eV),⁵⁰ α -C₃N₂ (0.27–0.34 eV)⁵¹ and so on; a more comprehensive comparison of these diffusion barriers is provided in Table 1.

The diffusion coefficients (D) for Li atoms along three different migration paths are calculated using the Arrhenius equation⁵² below:

$$D \approx \exp\left(-\frac{E_a}{k_B T}\right) \quad (3)$$

where E_a is the diffusion barrier, and k_B and T represent the Boltzmann constant (8.617×10^{-5} eV K⁻¹) and temperature, respectively. Consequently, at 301.8 K, the calculated diffusion coefficients for paths I, II, and III correspond to 9.74×10^{-6} , 1.34×10^{-5} and 9.85×10^{-4} , respectively. These results suggest that Li ions exhibit the fastest diffusion along path III, followed by path II and path I, confirming that Li prefers to migrate along the path with the lowest diffusion barrier.

3.4. Open circuit voltage and theoretical storage capacity of CKL

The storage capacity, an essential metric for anode performance, is primarily governed by the maximum concentration of Li atoms that can be adsorbed by the substrate. We initially examine the scenario for the adsorption of Li atoms to all H1 sites on the surface of CKL, as this is the most stable adsorption site. Once all the H1 sites are occupied, we gradually proceed to adsorb Li atoms on the H2 sites. However, after sequential

Table 1 Comparison of key properties (diffusion barrier (eV), capacity (mAh g⁻¹), and OCV (V)) between CKL and other 2D materials

Materials	Diffusion barrier (eV)	Capacity (mAh g ⁻¹)	OCV (V)
Net-C18 ⁶³	0.32–0.72	403	1.27
Thgraphene ⁶⁴	0.06–0.17	1489	0.5–0.6
HOP-graphene ²²	0.7–0.8	1338	0.42
QPHT-graphene ⁶⁵	0.36–0.83	837	0.66
popgraphene ⁶⁶	0.37–0.55	1487	0.45
C ₅₆₇₈ ⁵⁹	0.29–0.44	697	0.33
bco-C ₁₆ ⁶⁷	0.53–2.23	558	0.23
χ_3 -Borophene ⁶⁸	0.628	1240	1.09
2D-Borophene ⁶⁹	0.007–0.267	620	1.11
BC ₇ ⁷⁰	0.67–0.89	423.7	1.7
Si ₃ C ⁷¹	0.18–0.47	1394	0.58
α -CP ⁵⁷	0.11–0.5	1108.91	1.99
Ti ₂ C ⁷²	0.18	288	0.4
C ₃ N ⁷³	0.12	1092	0.45
α -C ₃ N ₂ ⁵¹	0.27–0.34	2791	0.62
B-Graphyne ⁷⁴	0.17–0.87	525	0.73
h-ALC ⁵⁶	0.78–0.85	739.61	1.38
BSi ⁵⁵	0.40–0.56	1374	1.31
B ₂ N ⁷⁵	1.38	1082	1.07
T-Be ₂ B ⁶	0.26–0.37	929	0.88
ABC-Si ₄ Al ₂ ⁷⁶	0.53–0.86	645	0.49
<i>t</i> -BN ⁷⁷	0.35–0.61	1620	0.479
α -SiN ⁷⁸	0.30	956	2.26
VO ₂ ⁷⁹	0.119	968	1.56
VS ₂ ⁸⁰	0.22	466	0.93
THD-C ⁸¹	0.49–0.64	1116.7	0.4
Me-C ₈ B ₅ ⁸²	0.66–0.79	2856	0.94
CKL	0.18–0.30	1115.7	1.3

adsorption of Li atoms up to 14 atoms on the H2 sites, we observe significant structural deformation in the material. The deformation becomes so severe that the system is no longer able to accommodate additional Li atoms on the H2 sites, indicating that further adsorption on these sites is not feasible. Thus, we consider the adsorption of Li atoms 3 Å above the H1 sites on the surface of CKL. The average adsorption energy is

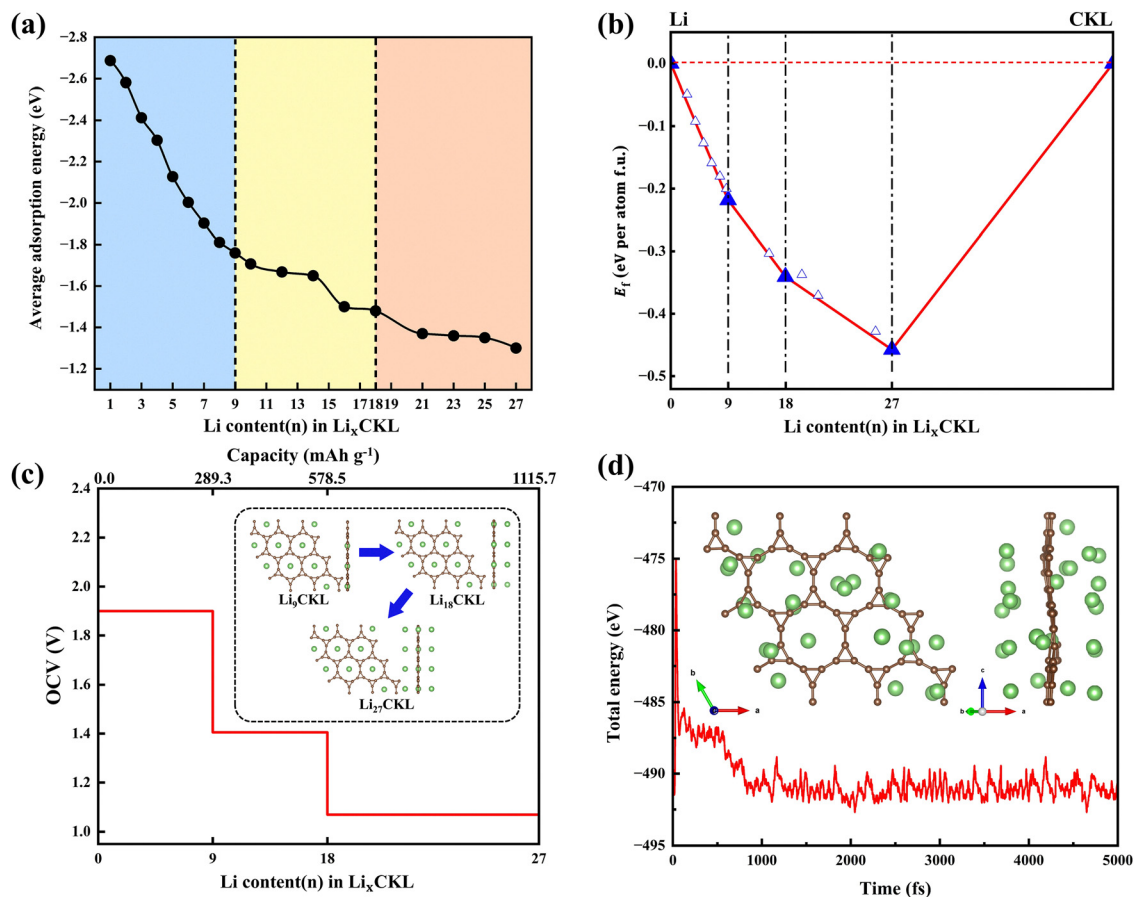


Fig. 6 (a) The average adsorption energy for adsorbing different quantities of lithium atoms. (b) The formation energy convex hull diagram of CKL. (c) The change of the OCV with the increase of the loaded lithium atoms. (d) AIMD simulation of Li₂₇CKL at 400 K, the insets display the top and side views of the final snapshot of Li₂₇CKL.

evaluated and displayed in Fig. 6(a), and the corresponding formula is expressed as follows:

$$E_{\text{ave}} = \frac{E_{n\text{Li}+\text{CKL}} - E_{\text{CKL}} - nE_{\text{Li}}}{n} \quad (4)$$

where $E_{n\text{Li}+\text{CKL}}$ is the energy of n Li atoms adsorbed on the $3 \times 3 \times 1$ supercell of the CKL substrate, while E_{CKL} and E_{Li} represent the energies of the $3 \times 3 \times 1$ supercell of CKL and a single Li atom, respectively. In the case of full adsorption at the most stable H1 site with 9 Li atoms on the surface of CKL, the average adsorption energy is -1.76 eV. For the second layer, where 9 Li atoms are adsorbed on the H1 site at a height of 3 Å from the surface, the calculated average adsorption energy is -1.48 eV. To explore the maximum amount of Li atoms that CKL can load, we further consider the third layer of adsorption, which follows the same strategy as the adsorption of the second layers. Finally, the CKL is loaded with a total of 27 Li atoms, and the average adsorption energy is -1.31 eV. However, extrapolating this trend indicates that the adsorption energies remain negative, without transitioning to positive values. While positive adsorption energy is not a necessary condition for the adsorption to be completed,⁵³ it necessitates a more thorough examination of the lithiation process. As illustrated in Fig. 7, 27

Li atoms are adsorbed onto the host material. It is observed that none of the Li atoms detach from the surface or push other lithium atoms away, indicating that the adsorption process is stable and does not lead to clustering or desorption.

The electronic properties of CKL upon adsorption of varying amounts of Li atoms (9, 18, and 27 Li) were analyzed through PDOS calculations (see Fig. 7). For low Li coverage (9 Li), the states near the Fermi level are predominantly derived from the C-pz orbitals, with minimal contribution from Li atoms. As the Li concentration increases to 18 Li, the states around the Fermi level intensify, particularly in the Li-p orbitals, suggesting a strengthening of the Li-C interaction, though still with weak hybridization. At the highest coverage (27 Li), the total DOS at the Fermi level significantly increases, primarily due to the enhanced Li-p orbital contribution, with a slight overlap with C-p orbitals. This weak orbital interaction suggests that Li remains predominantly ionic despite the increasing Li coverage, which is beneficial for reversible delithiation.

The ELF distributions further clarify the multilayer Li adsorption behavior of CKL. Fig. 7 shows that the Li atoms occupying the internal cavity (first layer) are surrounded by predominantly delocalized electron density, indicating weak Li-Li repulsion and a favorable electrostatic environment.

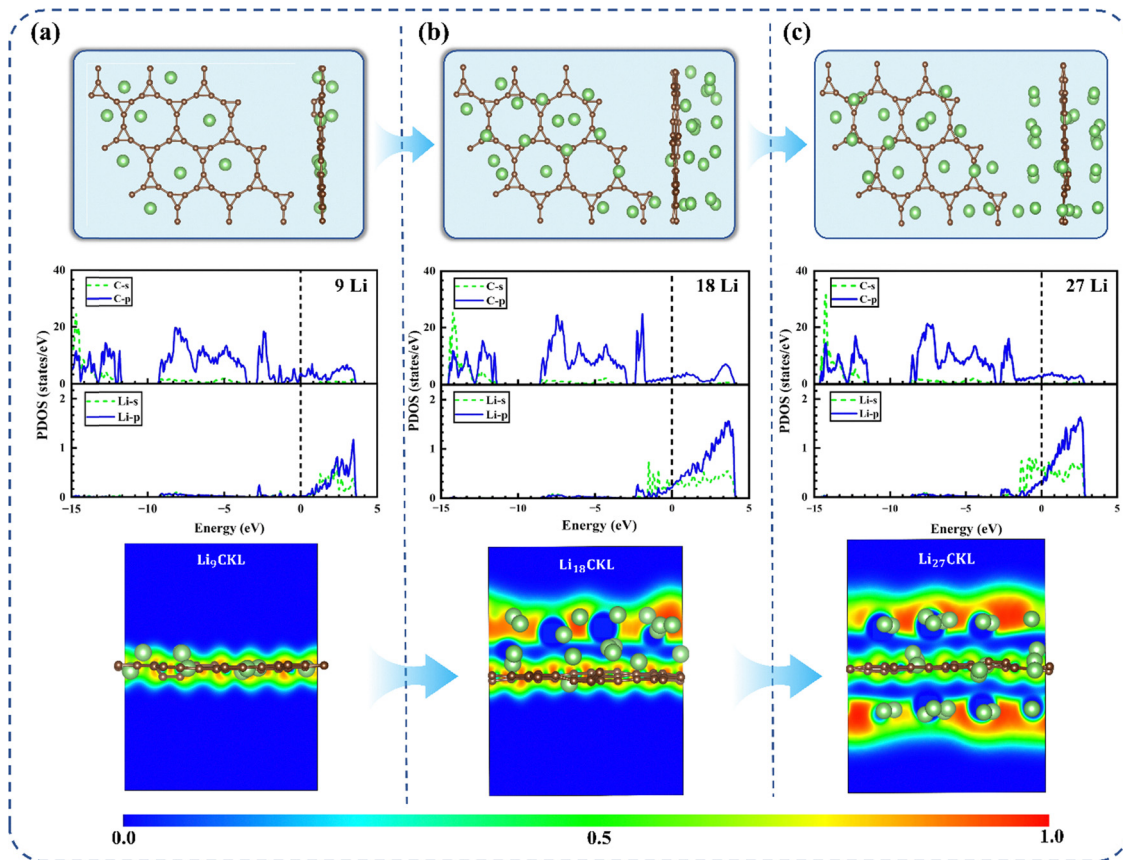


Fig. 7 The structure after relaxation, PDOS and ELF of the CKL monolayer with different numbers of lithium atoms (a) 9 Li, (b) 18 Li, (c) 27 Li.

This delocalization also allows additional Li atoms to be stably accommodated on both sides of the monolayer at approximately ± 3 Å (second and third layers). However, for the outer adsorption layers, the ELF maps reveal increasingly localized electron regions between adjacent Li atoms, especially when the Li coverage reaches the second surface layer. Such localization reflects enhanced Li–Li self-interaction that cannot be effectively screened by the CKL framework. When attempting further adsorption beyond these positions, the localized electrons between Li atoms become even more pronounced, signaling strong coulombic repulsion and thermodynamic instability.⁵⁴ Therefore, the maps of ELF confirm that CKL can host one Li layer inside its porous cavities and two additional layers on its two surfaces, while further adsorption is inhibited due to significant Li–Li repulsion.

To further assess whether the structures of adsorbed lithium are in a state with thermodynamic stability, we plot the convex hulls for the formation energy by using the formula below:

$$E_f = \frac{E_{\text{Li}_x\text{C}} - E_{\text{CKL}} - xE_{\text{Li}}}{x + 1} \quad (5)$$

where x represents the concentration of Li on the CKL monolayer. $E_{\text{Li}_x\text{C}}$ and E_{CKL} are the energies of x Li atoms adsorbed on CKL and pristine CKL, respectively. E_{Li} is the total energy of a Li atom. As presented in Fig. 6(b), the configurations

corresponding to Li_9CKL , Li_{18}CKL , and Li_{27}CKL are the most energetically favorable lithiated states.

OCV corresponds to the equilibrium voltage of the battery in the absence of an external circuit and reflects the thermodynamic driving force for Li insertion, making it a critical metric for assessing anode materials in lithium-ion batteries. The OCV associated with the charging process can be estimated from the following theoretical expression:

$$\text{OCV} = -\frac{E_{\text{Li}_{x_2}\text{CKL}} - E_{\text{Li}_{x_1}\text{CKL}} - (x_2 - x_1)E_{\text{Li}}}{e(x_2 - x_1)} \quad (6)$$

here $E_{\text{Li}_{x_1}\text{CKL}}$ and $E_{\text{Li}_{x_2}\text{CKL}}$ are the total energies of Li_xCKL at two adjacent concentrations of x_1 and x_2 ($x_1 \leq x \leq x_2$), respectively. E_{Li} represents the energy of a single Li atom. We only calculate the open-circuit voltage for structures with thermodynamic stability. As shown in Fig. 6(c), the voltage profile exhibits three distinct plateaus corresponding to the stable configurations on the convex-hull curve (Li_9CKL , Li_{18}CKL , and Li_{27}CKL). The initial voltage is around 1.90 V, corresponding to Li adsorption on the most energetically favorable H1 sites. As the Li concentration increases, the OCV gradually decreases to approximately 1.41 V, indicating a progressive weakening of the Li–substrate interaction accompanied by an increasing contribution from Li–Li repulsion. With further lithiation up to Li_{27}CKL , the OCV continues to decrease to approximately 1.06 V. The average OCV of CKL for all steady

states is 1.3 V. Generally, a relatively high OCV can help reduce the growth of dendrites on the electrode surface.⁵⁵ The calculated average OCV of CKL is significantly lower than other 2D anode materials such as h-ALC (1.38 V),⁵⁶ α -CP (1.99 V).⁵⁷ Such a moderate OCV is high enough to mitigate the associated risk of excessive Li plating, while remaining sufficiently low to preserve favorable cell output.

We then calculated the maximum theoretical storage capacity of the system using

$$C = \frac{x_{\max} F}{M} \quad (7)$$

where $x_{\max}$ represents the maximum concentration of Li atoms. F and M are the Faraday constant (26 801 Ah mol) and molecular weight of CKL, respectively. Therefore, the theoretical capacity of CKL as an anode material can reach up to 1115.7 mAh g⁻¹, which surpasses that of many 2D known anode materials such as Ψ -graphene (372 mAh g⁻¹),²⁶ phagraphene (558 mAh g⁻¹),⁵⁸ C₅₆₇₈ (697 mAh g⁻¹),⁵⁹ and close to k -B₂C₃ (930 mAh g⁻¹)⁶⁰ and silence (954 mAh g⁻¹)⁶¹ (Table 1). Nevertheless, the calculation of the theoretical capacity is based on an ideal vacuum model, which neglects the effects of defects, interfaces, electrolytes, and other electrode-level complexities. It may somewhat overestimate the practical capacity. Accordingly, investigating the practical performance of CKL under realistic operating conditions will be an important direction for future research.

In practical applications, the thermal stability of electrode materials is a key factor for ensuring battery safety and lifetime. Commercial batteries typically operate within -20 to 55 °C ($\approx$ 253 to 333 K), while carbon-based anodes often face risks of SEI decomposition or structural instability at 80–120 °C ($\approx$ 353 to 393 K).⁶² To rigorously test the thermal stability of CKL under extreme conditions, AIMD simulations were performed at 400 K for both 12 Li atom adsorbed configurations. The results reveal only slight structural fluctuations throughout

the simulation, with no observable bond breaking or lattice reconstruction. These observations confirm the remarkable thermal stability of the CKL monolayer. To further examine structural reversibility, all Li atoms were removed from the fully adsorbed configuration and an additional AIMD simulation was performed at 400 K, as illustrated in Fig. 8. The initially distorted CKL framework rapidly restores its initial configuration, confirming that the CKL monolayer can reliably withstand repeated the actual battery charging and discharging processes while maintaining structural stability.

4. Conclusions

This work investigates the CKL monolayer as a potential anode material for Li-ion batteries using first-principles calculations. CKL is composed of three-membered and twelve-membered rings, and its structural stability is confirmed through phonon dispersion and AIMD simulations. Notably, the framework maintains its integrity not only in its pristine structure but also after accommodating a full loading of Li atoms. The material exhibits a stable adsorption site for a single Li atom at the center of the twelve-membered carbon ring (-2.68 eV), which originates from significant charge transfer (0.943e). The analysis of electronic properties reveals that the adsorption of Li enhances the density of states near the Fermi level, ensuring high metallic conductivity and efficient ion transport. Additionally, upon Li adsorption, charge transfer from Li to CKL slightly enhances the density of states near the Fermi level, and the structure still maintains its metallic nature owing to the significant interaction between the Li atoms and the carbon framework. Besides, CKL shows an extremely low diffusion barrier (0.18 eV) for Li, suggesting fast Li-ion diffusion, crucial for high-rate performance. The material can accommodate up to 27 Li atoms per supercell, offering a high theoretical storage capacity of 1115.7 mAh g⁻¹, surpassing many other 2D anode materials. To further assess the structural response of CKL under repeated charging and discharging processes, AIMD simulations were performed under both full adsorption and desorption conditions. The CKL monolayer preserves its structural integrity, exhibiting only slight deformation and a pronounced tendency to recover its original framework, which collectively indicates excellent structural reversibility and cycling stability during lithiation. Overall, the intrinsic metallic behavior, robust structural stability, low diffusion barriers, and high theoretical capacity of CKL, combined with its suitable OCV, make it a promising anode material for LIBs. These findings not only highlight the potential of CKL for well-balanced energy storage performance but also pave the way for further experimental studies and the eventual commercialization of advanced and durable battery materials.

Author contributions

Haoxuan Chen: data curation; investigation; writing – original draft; conceptualization; formal analysis; methodology; software;

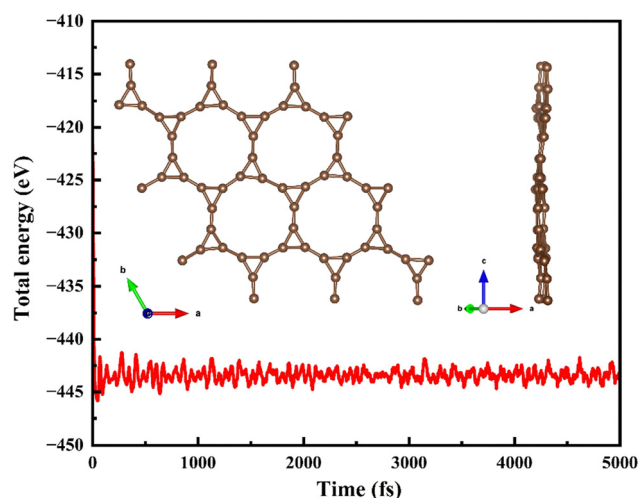


Fig. 8 AIMD simulation of CKL after Li removal at 400 K; the inset displays the top and side views of the structure of CKL before and after dynamics simulations.

writing – review & editing; visualization. Zhenjie Zhao: formal analysis; methodology; software; visualization. Zihao Wang: formal analysis; methodology; software; visualization. Huadong Zeng: conceptualization; funding acquisition; resources; supervision; investigation; formal analysis; methodology; software; visualization; writing – review & editing. Xinlu Cheng: formal analysis; funding acquisition; resources; supervision; investigation; software.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Supplementary information (SI) includes AIMD at 500 K for 12 ps and Projected crystal orbital Hamilton population (pCOHP) of Li–C interactions at H1 site. See DOI: <https://doi.org/10.1039/d6cp00574h>.

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